



Final Technical Report

Project Title

Digital Assistant for Classroom Management

Submitted By

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1. Introduction

The Digital Assistant for Classroom Management is a Flask-based web application built to help faculty manage student academic data, monitor performance, review department-level trends, and use AI-assisted risk analysis from a single dashboard. The system was developed as a practical academic administration tool with emphasis on usability, responsiveness, and robust error handling.

2. Problem Statement

Faculty teams often manage student data across multiple spreadsheets and disconnected tools. That approach makes it difficult to track attendance, semester performance, risk levels, and department-level insights efficiently. The project addresses this issue by centralizing student management, analytics, reporting, and AI-supported interpretation in one web application.

3. Objectives

- Build a web-based classroom management assistant for faculty.
- Provide structured student profile and academic record management.
- Generate reports and charts for student and department review.
- Offer AI-assisted performance and risk analysis.
- Improve usability through professional UI refinement and clear feedback states.
- Handle invalid input and service failures gracefully without crashing.

4. Technology Stack

- Backend: Python, Flask
- Frontend: HTML, CSS, JavaScript, Jinja2 templates
- Data Storage: CSV files, SQLite, JSON
- Reporting: ReportLab PDF generation
- Charts: Matplotlib
- Data Processing / AI Logic: pandas, scikit-learn
- Deployment Target: Render

5. System Architecture

The application uses a modular Flask structure:

- ``app.py`` is the entry point and exposes the Flask application object.
- ``classroom_app/__init__.py`` creates the app, registers blueprints, initializes the database, and configures global error handling.
- ``classroom_app/blueprints/auth.py`` handles login, logout, and route protection.
- ``classroom_app/blueprints/pages.py`` handles user-facing pages such as dashboard, students, feedback, AI logs, and form workflows.
- ``classroom_app/blueprints/api.py`` exposes API endpoints for AI analysis and AI log retrieval.
- ``classroom_app/services/`` contains data, chart, logging, and student-management logic.
- ``templates/`` contains the UI pages.
- ``static/`` contains CSS, JavaScript, generated charts, and images.

6. Major Modules

6.1 Authentication Module

Faculty login is handled using session-based authentication. Protected pages require successful login before access is granted.

6.2 Student Management Module

The system supports adding, editing, deleting, filtering, sorting, and exporting student records. Validation is applied to reduce incorrect or incomplete submissions.

6.3 Student Profile and Analytics Module

Each student profile presents semester-wise academic details, attendance, generated charts, and AI-supported insights.

6.4 Department and Leaderboard Module

Department pages summarize student information at a higher level and support report export. A leaderboard view highlights comparative performance.

6.5 AI Analysis Module

The AI module processes student data and returns performance-oriented analysis. The API layer includes timeout protection and safe error responses to prevent crashes during long-running or failed requests.

6.6 Feedback and Activity Logging Module

The system stores feedback entries and records major user actions such as login, AI analysis, export operations, and student updates.

7. Data and File Storage

The project uses multiple storage formats:

- `database/students.csv` for student records
- `database/courses.csv` and `database/semesters.csv` for academic reference data
- `database/app.db` for SQLite-based AI log storage
- `data/feedback.json` for feedback entries
- `activity\_log.txt` for activity history
- `static/charts/` for generated chart outputs
- `static/images/students/` for uploaded student images

8. UI and UX Refinement

Phase 3 focused on improving the presentation and usability of the application:

- Shared buttons, forms, and input styles were refined for consistent appearance.
- Layout spacing and component behavior were improved for a more professional UI.
- Loading states were added for long-running actions such as AI processing, filtering, export tasks, and profile loading.
- Clear success and error messages were added to improve usability.

9. Robustness and Stress Handling

The system was updated to fail safely instead of crashing:

- Invalid form input is validated before processing.
- File upload issues are handled with user-facing error messages.
- AI processing is wrapped with timeout protection.
- API failures return clear JSON error responses.

- Unhandled server errors are caught by a global Flask error handler and rendered through a user-friendly error page.

This directly supports the evaluation criterion requiring graceful handling of invalid input and API timeouts.

10. Deployment

The application is prepared for production deployment on Render.

Recommended Render configuration:

- Build command: ``pip install -r requirements.txt``
- Start command: ``gunicorn --bind 0.0.0.0:$PORT app:app``
- Environment variable: ``SECRET_KEY=<secure-random-value>``

Because the app uses SQLite and local files, the deployment must account for persistent storage limitations on cloud platforms with ephemeral filesystems.

11. Testing and Verification

The project was verified through functional testing of major workflows:

- Login and logout
- Student add, edit, delete, and filtering
- Feedback submission
- AI analysis request and timeout handling
- AI log viewing
- CSV/PDF export flows
- Error page rendering for invalid routes

Stress checks included invalid inputs, missing records, and service failure scenarios to confirm that the system shows clear feedback instead of exposing raw exceptions.

11.1 Website Screenshots

This report can include screenshots of the deployed website or local application pages such as:

- Login page
- Dashboard page
- Student management page
- Student profile page
- AI logs page

Screenshot files placed in `docs/screenshots/` are included automatically in the generated PDF.

12. Outcomes

The completed system delivers:

- A centralized classroom management workflow
- Better visibility into academic performance
- AI-supported student risk interpretation
- Export-ready reporting
- Cleaner UI and stronger user feedback
- Safer behavior under invalid input and failure conditions

13. Limitations

- The app currently relies on local-file storage and SQLite.
- Role-based access is limited.
- Production persistence needs extra setup on some hosting platforms.
- AI logic can be improved further with more advanced models or real external services.

14. Future Enhancements

- Add role-based access control
- Move data storage to a managed production database
- Add email or notification support
- Improve AI recommendations with richer academic history
- Add more analytics dashboards and downloadable reports

15. Conclusion

The Digital Assistant for Classroom Management successfully meets the project goals by combining data management, reporting, AI-assisted analysis, and usability improvements in one faculty-focused application. The final version is more polished, more resilient, and better prepared for live deployment and academic demonstration.

Embedded Website Screenshots



**Bannari Amman
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Technology**
Digital Classroom
Assistant

Dashboard

Student Management

Leaderboard

Activity

AI Logs

Light Mode

Logout

Courses

Feedback

Admin Activity Log

Audited actions for student management workflows.

Timestamp	User	Action	Details
2026-03-12 08:52	admin	Faculty Logged In	Successful login
2026-03-12 08:44	admin	Faculty Logged In	Successful login
2026-03-12 08:41	admin	Faculty Logged In	Successful login
2026-03-12 08:39	admin	Faculty Logged In	Successful login
2026-03-12 08:38	admin	Viewed Student Profile	Poornesh (7376232AD209)
2026-03-12 08:38	admin	Exported PDF For Student	Poornesh
2026-03-12 08:38	admin	Viewed Student Profile	Poornesh (7376232AD209)
2026-03-12 08:37	admin	Faculty Logged In	Successful login
2026-03-12 08:35	admin	Faculty Logged Out	Session ended

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Activities



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Digital Classroom Assistant

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After Login



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Assistant

Dashboard

Student Management

Leaderboard

Activity

AI Logs

Dark Mode

Logout

Courses

Feedback

AI Analysis Logs

Stored responses from the student risk analyzer. Demo data is seeded automatically when no logs exist.

Student Name

Rows

Apply Filters

{ } Open JSON API

ID	Student	Risk Level	Performance Score	Timestamp
5	Aarav Kumar	High Risk	9.14	2026-03-12 06:30:49
4	Aarav Kumar	High Risk	9.14	2026-03-12 06:30:49
3	Rahul Nair	High	4.8	2026-03-07 04:26:29
2	Diya Mishra	Medium	6.9	2026-03-07 04:26:29
1	Aarav Kumar	Low	8.7	2026-03-07 04:26:29

Ai Logs



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Digital Classroom Assistant

AI Powered Academic Performance Monitoring

Before Login



**Bannari Amman
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Assistant

Dashboard

Student Management

Leaderboard

Activity

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Courses

Feedback

Light Mode

Logout

Course Management

Create, search, edit, and manage courses.

Add Course

Course Code

Course Name

Add Course

Search Course

Search by code or name

Search

All Courses

Course Code	Course Name	Actions
22AI037	TIME SERIES ANALYSIS AND FORECASTING	EditDelete

Courses



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Activity

AI Logs

Courses

Feedback

Light Mode

Logout

Feedback

Share your experience to help improve the Digital Classroom Assistant.

Name

Email

Feedback

Rating

Select Rating

Submit Feedback

Back to Dashboard

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Feedback

**Overall Topper**

Aditya Patel

Computer Technology
500 marks

Rank	Name	Department	Sem6 Total
1	Aditya Patel	Computer Technology	500
2	Poornesh	Artificial Intelligence and Data Science	500
3	Reyansh Iyer	Electronics & Instrumentation Engineering	500
4	Rohan Sharma	Computer Science & Engineering	500
5	Siddharth Reddy	Artificial Intelligence and Machine Learning	499

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Leader Board

Empowering Digital Learning

Smart classroom management for modern institutions

AI

CLD

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Faculty Login
Digital Classroom Assistant

Username

Enter username

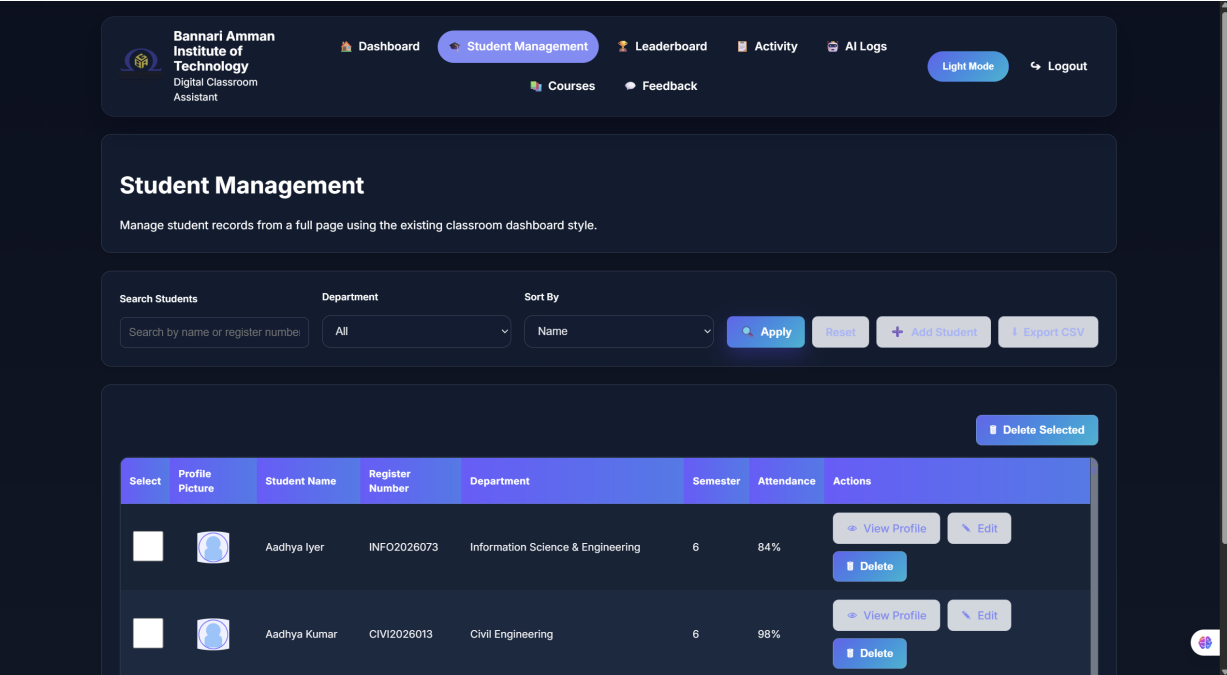
Password

Enter password

Login

Login Page

Page 9



Student Management